

A Heavy-Tail Obstruction to Arbitrary Santaló Gradient-Flow Curves

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Source Question

Cordero-Erausquin, Fradelizi, and Langharst ask in Section 5.2 of arXiv:2409.05541 whether the time-dependent gradient-flow equation

$$s'(t) = -\frac{p}{2q} \nabla_z Q(t, s(t))$$

has at least one sufficiently smooth solution on $\mathbb{R}_+$ with $\lim_{t \rightarrow \infty} s(t) = 0$. Here $0 < p < 1$, $q = p/(p-1) < 0$, and

$$Q(t, z) = \log \int_{\mathbb{R}^n} \mathcal{L}_p(\tau_z f_t)(x) dx.$$

The source passage is shown in Figure 1.

The previous packet

`solutions/partial/2409.05541_santaló_gradient_flow_regular_existence/`

proves existence for bounded compactly supported nonzero data under the Fokker-Planck flow. The point of this note is complementary: the literal arbitrary-nonnegative-function formulation is not true without a domain hypothesis on the double Laplace transform.

Idea of the Counterexample

If f has two-sided polynomial tails, then positive heat or Fokker-Planck time smooths f but does not create nonzero exponential moments. Thus

$$H_t(\xi) = \int_{\mathbb{R}} f_t(y)^{1/p} e^{\xi y} dy$$

is infinite for every $\xi \neq 0$. Since $q < 0$, the p -Laplace transform H_t^q is then zero almost everywhere. The integral defining Q is zero, so $Q = \log 0 = -\infty$ and the proposed vector field is nowhere defined.

The Counterexample

Theorem 1. Fix $0 < p < 1$ and $q = p/(p-1)$. In dimension one, let

$$f(x) = \frac{1}{1+x^2}.$$

Let f_t be either the heat evolution $E_t f$ or the Fokker-Planck evolution $P_t f$ at any time $t > 0$. Then

$$\int_{\mathbb{R}} \mathcal{L}_p(\tau_z f_t)(\xi) d\xi = 0$$

for every $z \in \mathbb{R}$. Consequently

$$Q(t, z) = \log \int_{\mathbb{R}} \mathcal{L}_p(\tau_z f_t) = -\infty$$

for every $t > 0$ and $z \in \mathbb{R}$, and the gradient-flow equation

$$s'(t) = -\frac{p}{2q} \nabla_z Q(t, s(t))$$

has no sufficiently smooth solution in the usual real-valued sense.

Proof. First consider the heat flow

$$E_t f(y) = (2\pi t)^{-1/2} \int_{\mathbb{R}} \frac{e^{-(y-u)^2/(2t)}}{1+u^2} du.$$

For every fixed $t > 0$ there are constants $a_t, b_t > 0$ such that

$$\frac{a_t}{1+y^2} \leq E_t f(y) \leq \frac{b_t}{1+y^2}, \quad y \in \mathbb{R}. \quad (1)$$

For the lower bound, when y is large positive integrate over $u \in [y, y+1]$, and when y is large negative integrate over $u \in [y-1, y]$; the remaining compact set is absorbed into the constant. For the upper bound, use

$$1+y^2 \leq 2(1+u^2)(1+(y-u)^2)$$

inside the Gaussian convolution.

It follows from (1) that

$$E_t f(y)^{1/p} \asymp (1+y^2)^{-1/p}.$$

Since $p < 1$, the function $(1+y^2)^{-1/p}$ is integrable, but it has no nonzero two-sided exponential moment. Therefore

$$H_t(\xi) := \int_{\mathbb{R}} E_t f(y)^{1/p} e^{\xi y} dy$$

is finite and positive at $\xi = 0$, and is infinite for every $\xi \neq 0$. Thus, with $q < 0$,

$$\mathcal{L}_p(E_t f)(\xi) = H_t(\xi)^q$$

is zero for every $\xi \neq 0$. The value at the single point $\xi = 0$ is finite, but it has no effect on Lebesgue integration. Using the source convention

$$\mathcal{L}_p(\tau_z g)(\xi) = \mathcal{L}_p(g)(\xi) e^{qz\xi},$$

we obtain

$$\int_{\mathbb{R}} \mathcal{L}_p(\tau_z E_t f)(\xi) d\xi = \int_{\mathbb{R}} \mathcal{L}_p(E_t f)(\xi) e^{qz\xi} d\xi = 0$$

for every $z \in \mathbb{R}$.

The Fokker-Planck case is the same after the standard heat/Fokker-Planck change of variables used in the source paper:

$$P_t f(y) = e^{t/2} E_{e^t - 1} f(e^{t/2} y).$$

The comparability (1) is preserved under this scaling, so the same argument gives

$$\int_{\mathbb{R}} \mathcal{L}_p(\tau_z P_t f)(\xi) d\xi = 0$$

for every $t > 0$ and $z \in \mathbb{R}$.

Hence, for either semigroup, $Q(t, z) = \log 0 = -\infty$ everywhere. In particular Q is not a finite differentiable function and $\nabla_z Q(t, z)$ is not defined. No C^1 curve can satisfy the displayed gradient-flow ODE in the ordinary sense. $\square$

Scope

This is a counterexample only to the literal all-nonnegative-data reading of the source question. It does not contradict the regular positive theorem in the companion packet. Instead, it shows that some finite-domain or Gaussian-tail type hypothesis is necessary before the gradient-flow question is well posed.

Novelty Check

The source paper itself discusses heavy-tail examples where $\mathcal{L}_p(f)$ vanishes away from the origin. The extra observation here is that, for polynomial tails, this obstruction persists after every positive heat or Fokker-Planck time. Cheap run indexes and a bounded web check on July 19, 2026 found no duplicate packet or later exact answer to the Section 5.2 gradient-flow domain question.

References

1. D. Cordero-Erausquin, M. Fradelizi, and D. Langharst, *On a Santaló point for Nakamura-Tsuji's Laplace transform inequality*, arXiv:2409.05541; Forum of Mathematics, Sigma 13 (2025), e125.

It would be interesting to know if, after translating f , the single point zero curve belongs to $S_p(f)$ or if the curves stay in a compact region that can be described. Actually, we don't even know if for the volume product, i.e. when $p = 0$, $t \mapsto \int f_t \int (f_t)^\circ$ increases when f or f° has barycenter 0, say. Note also that monotonicity ensures that $\int f_t \int (\tau_{s(t)} f)^\circ \leq (2\pi)^n$, and so $s(t)$ has to belong to the so-called 'Santaló region' [27, 16] of f_t .

For instance, in the trivial case where $f(x) = e^{-|x-a|^2/2}$ is a Gaussian centered at $a \in \mathbb{R}^n$, then it is readily checked that $s(t) = e^{-t/2}a$ is the only element of $S_p(f)$ for all $p \in [0, 1)$.

One can also look at the subset of curves in $S_p(f)$ that make our argument work with fixed $p \in (0, 1)$. In the proof of (32) in Theorem 3.1, given in Section 3.4, we see that establishing the monotonicity in t of $M_p(f_t)$ (assuming f belongs to the class $\mathcal{C}$ defined in (40)) amounted to the fact that

$$\begin{aligned} \alpha'(t) &= \partial_t Q(t, s_p(f_t)) + \nabla_z Q(t, s_p(f_t)) \cdot \partial_t s_p(f_t) \\ &\geq \frac{p}{2q} |\nabla_z Q(t, s_p(f_t))|^2 + \nabla_z Q(t, s_p(f_t)) \cdot \partial_t s_p(f_t) \\ &= 0, \end{aligned}$$

where $Q(t, z) = \log L(\mathcal{L}_p(f_t))(qz) = \log \int \mathcal{L}_p(\tau_x(f_t))$ is convex in z , and $\alpha(t) = Q(t, s_p(f_t))$. In particular, our choice of $z = s_p(f_t)$ ensured that

$$|\nabla_z Q(t, s_p(f_t))|^2 = 0 = \nabla_z Q(t, s_p(f_t)) \cdot \partial_t s_p(f_t).$$

But we see that we only need the curve to satisfy

$$\frac{p}{2q} |\nabla_z Q(t, s(t))|^2 + \nabla_z Q(t, s(t)) \cdot s'(t) \geq 0, \quad (58)$$

which leads to the subset $\tilde{S}_p(t) \subset S_p(f)$

$$\tilde{S}_p(f) = \{s \in C^1(\mathbb{R}^+, \mathbb{R}^n) ; \lim_{t \rightarrow \infty} s(t) = 0 \text{ and (58) holds at every } t > 0, z \in \mathbb{R}^n\}.$$

A natural candidate, besides our p -Santaló points $s_p(f_t)$ themselves, for which both terms are zero, would be a curve satisfying the equation

$$s'(t) = -\frac{p}{2q} \nabla_z Q(t, s(t)) = \frac{p}{2} \int_{\mathbb{R}^n} x \frac{\mathcal{L}_p(\tau_{s(t)} f_t)(x)}{\int_{\mathbb{R}^n} \mathcal{L}_p(\tau_{s(t)} f_t)(x) dx} dx. \quad (59)$$

We leave open the following natural question: does this time-dependent gradient flow equation have at least one (smooth enough) solution on $\mathbb{R}^+$ with $\lim_{t \rightarrow \infty} s(t) = 0$? In the case $p = 0$, the equation becomes

$$s'(t) = \frac{1}{2} \nabla \log L((f_t)^\square)(s(t)).$$

In all cases, one probably needs to understand the time dependent log-Laplace transform $Q(t, z)$.

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Figure 1: Open question on Santaló gradient-flow curves, source PDF page 33.